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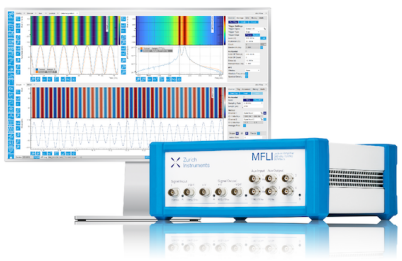
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Thermo-physical properties of LiH at high pressures by *ab initio* calculations

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First principles calculations have been carried out to analyze structural stability and to determine the equation of state and elastic constants of LiH as a function of pressure. The comparison of total energies of B1 and B2 structures determined as a function of compression suggests the B1 → B2 transition at ~ 327 GPa. Various physical quantities including zero pressure equilibrium volume, bulk modulus, pressure derivative of bulk modulus, Debye temperature, bulk sound speed, Hugoniot parameter '*s*' and Gruneisen parameter have been derived. All these physical quantities compare well with the available experimental data. The single crystal elastic constants have been evaluated up to the B1→B2 transition pressure. © *2011 American Institute of Physics*. [doi:10.1063/1.3590150]

I. INTRODUCTION

Lithium hydrides (LiH, LiD), simple ionic compounds having NaCl structure at ambient condition, have attracted much attention because of their use in nuclear industry and for potential energy production. An extensive amount of experimental and theoretical work has already been reported in past few decades[1–16] to understand their solid state properties. Many of these studies are also related to high pressure behavior of LiH, which include analysis of its phase stability,[1–10] determination of equation of state,[2,3,7–12] understanding the effect of pressure on electronic charge density,[12] band structure[8,12,13] and melting point[15] etc. As far as structural stability of this compound under high pressure is concerned, widely differing results have been reported in literature.[1,2,16–19] On the basis of psuedopotential calculations within local density approximation (LDA), Martins[16] predicted NaCl type (B1 phase) to CsCl type (B2 phase) structural phase transition at ~ 450 GPa, whereas, theoretical calculations by Gou *et al.*[17] using ionic overlap compression model found this transition to occur at ~ 85 GPa. In contrast to this, the x-ray diffraction measurements in DAC carried out by Loubeyre *et al.*[10] do not show any phase transition up to 94 GPa. Subsequent theoretical studies by Ahuja *et al.*[1] using full potential LMTO and Wang *et al.*[2] using full potential LAPW (WIEN97 package) put this transition at ~400 GPa and 313 GPa, respectively. However, using pseudopotential technique Zhang *et al.*[18] and Zurek *et al.*[19] found the transition pressure as ~ 200 GPa and 300 GPa, respectively. In addition to the direct total energy calculations for determining the structural stability, one can also get this information from the temperature or pressure variation of elastic constants. Any softening in them may be related to a phase transition or vanishing of some elastic constants may indicate mechanical instability of a material. In the present study, we report *ab-initio* theoretical determination of high pressure equation of state and pressure dependent elastic constants of LiH. There are only a few experimental and theoretical studies on elastic properties of this compound.[14,18,20–22] Further, the knowledge of these high pressure elastic properties may be more useful for applications in nuclear industry.

II. Theoretical Procedure

To analyze the structural stability under hydrostatic compression, we have computed the total energy for B1 and B2 phases of LiH at 0K as a function of volume up to about 450 GPa. In order to derive the theoretical isotherm, a polynomial fit was used to determine the pressure. The zero pressure bulk modulus (B_0) and its pressure derivative at zero pressure (B_0') were obtained by fitting the P-V data to the third order Birch-Murnaghan equation of state.[23] The 300K isotherm is derived by adding lattice thermal corrections using the procedure followed by Wallace[24] and Godwal and Jeanloz.[25]

For the LiH single crystal having cubic structure, there are only three independent elastic constants, C_{11}, C_{12} and C_{44} or equivalently three independent elastic moduli namely bulk modulus (B) and two shear moduli $C' = (C_{11} - C_{12})/2$ and C_{44}. To determine the shear moduli C' and C_{44} at a particular volume, the total energy is calculated as a function of different kinds of distortions[26,27] with the dependence of the total energy of the strained lattice on the deformation tensor as expressed in Ref. 28:

$$E_c(V, \underline{\varepsilon}) = E_c(V, 0) + \frac{1}{2} V \sum_i \sum_j C_{ij} e_i e_j. \qquad (1)$$

The shear elastic moduli are then calculated from the second order derivative of the total energy with respect to strain e_i (with $i = 1$ to 6). Further, we have used these single crystal elastic constants to determine the isotropic elastic properties of the polycrystalline LiH containing randomly oriented crystallites through the two well-known averaging schemes, one based on strain continuity across the grain boundaries suggested by Voiget,[29] and the other based on stress

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continuity across the grain boundaries by Reuss.[30] For LiH having cubic structure, the polycrystalline moduli can be expressed as:

$$B = \frac{1}{3}(C_{11} + 2C_{12}) \tag{2}$$

$$G_V = \frac{1}{5}(C_{11} - C_{12} + 3C_{44}) \tag{3}$$

$$G_R = \frac{5C_{44}(C_{11} - C_{12})}{3(C_{11} - C_{12}) + 4C_{44}}, \tag{4}$$

where $B = B_V = B_R$ with suffix V and R representing the Voiget and Reuss averaging, respectively.

The total energy calculations required for structural stability analysis, determination of 0 K isotherm and elastic moduli have been performed using WIEN2K package[31] based on FP-LAPW method. The exchange correlation part of the total energy is evaluated using generalized gradient approximation (GGA).[32] In order to test the applicability of different exchange-correlation functionals, some calculations have also been done with LDA. It is found that the equation of state parameters obtained from GGA calculations are in better agreement with the experimental data (see below). A grid of 5000 k points has been used for sampling the Brillouin zone. The plane wave cutoff parameter $R_{MT}\,K_{MAX}$ is fixed at seven with muffin tin radius R_{MT} chosen to be 1.3 a.u. for Li and 1.0 a.u. for H. The variation in R_{MT} around these values does not affect the results. The self-consistent cycle was run until the energy convergence criterion of 0.1 mRy was met.

III. RESULTS AND DISCUSSIONS

In Table I, we compare various physical quantities for B1 phase derived from our calculations with the experimental data and theoretical values available from other sources.[5–8,10,11,20–22] The calculated zero pressure volume, bulk modulus and its pressure derivative at zero pressure volume obtained from GGA (LDA) calculations agree within 1.4(8.3), 1.4(19.4) and 2.3(1.4) %, respectively with experimental values.[10,20] The experimental data agree better with the GGA results. This is consistent with the theoretical study by Lebegue *et al.*[3] using projector – augmented – wave method under GGA. Further, as depicted in the Table I, our theoretical zero pressure bulk modulus of 34.2 GPa from GGA and 40.3 GPa from LDA calculations at 0 K agrees well with GGA and LDA values of 36.2 GPa and 40.5 GPa reported by Barrera *et al.*[5] from pseudopotential calculations at 0 K. However, the room temperature value of B_0 by these authors under quasiharmonic approximation is $\sim$29 GPa for GGA and $\sim$33 GPa for LDA, suggesting that the value determined from LDA is closer to the experiment (32.2 GPa). We find that, if instead of quasiharmonic approximation, the experimentally measured temperature derivative of bulk modulus (~ -0.0083 GPa/K)[20] is used, the 0 K value reported by Barrera *et al.*,[5] becomes $\sim$34 GPa for GGA and $\sim$38 GPa for LDA at the room temperature. Interestingly, as shown in the Table I, we find that though, the zero pressure volume and bulk modulus are different for GGA and LDA, the pressure derivative of bulk modulus is almost the same

TABLE I. Comparison of theoretical and experimental values of various physical properties of LiH at zero pressure.

Properties	Present Theoretical Work[a]	Theoretical (Other sources)	Experimental
$V_{0(300K)}$ (Å^3/ formula unit)			
GGA	17.26	17.76[b],16.15[c], (15.18–16.24)[d], 15.24[e]	17.02[h], 16.85[i]
LDA	15.60	16.46[b], (14.87–15.18)[d], 16.0[f]	
B_0 (GPa)			
GGA	34.2	36.2[b], 36.07[c], (34.03–39.99)[d], 34.30[e]	32.2[h], 33.74[j],
LDA	40.3	40.5[b], 36.6[f], (41.33–42.8)[d]	32.35[l]
B_0'			
GGA	3.61	3.49[c], 3.40[e]	3.53[h]
LDA	3.58	3.40[f]	(3.8 ± 0.15)[l]
C_b (km/s)	6.48	–	6.43[j]
s	1.15	–	1.16[m]
γ	1.30	0.85[f]	1.2[j]
C_{11}(GPa)	77.0	82.7[e], 69.28[g]	74.06[k], 72.83[j], 67.2[l]
C_{12}(GPa)	12.7	10.7[e], 17.14[g]	14.2[j], 14.93[l]
C_{44} (GPa)	48.2	52.5[e], 58.25[g]	48.43[k] 48.0[j], 46.37[l]
θ_0 (K)	828	1246[f]	810[n]

[a]All the quantities are at zero pressure and zero temperature except the volume which is at zero pressure and room temperature.
[b]Zero pressure equilibrium volume at room temperature and bulk modulus at 0 K from pseudopotential calculations within GGA and LDA (Ref. 5).
[c]Values at 0 K from FP-LAPW calculations within GGA(Ref. 6).
[d]Values at 0 K from PBE(NC), PW(NC), PW(US) calculations within GGA and PZ(NC), PZ(US) calculations within LDA (Ref. 7).
[e]Values from *ab-initio* pseudopotential plane wave calculations under GGA (Ref. 18).
[f]Values at 0 K from APW calculations within LDA (Ref. 8).
[g]Values at 0 K from Hartree-Fock calculations (Ref. 22).
[h,i]Experimental data at room temperature from high pressure x-ray diffraction study (Ref. 10 and Ref. 11).
[j,k]Experimental values at ambient pressure and, 140 K and 60 K determined from measurements of velocity of ultrasonic waves (Ref. 20).
[l]Experimental values at ambient pressure and room temperature determined from measurements of velocity of ultrasonic waves (Ref. 21).
[m]Experimental value from shock wave experiments (Ref. 35).
[n]Value determined from experimentally measured specific heat (Ref. 36).

for both the approximations. This is consistent with the argument by Knuc and Syassen[33] that the third order derivative of total energy with respect to volume is nearly independent of the approximation used for the exchange-correlation functional. Additionally, we have also made contact with the available shock data by evaluating the parameters for the linear shock velocity (U_s) versus particle velocity (U_p) relation:

$$U_s = C_b + sU_p \tag{5}$$

here, the constant C_b. is approximately related to the bulk modulus and the s parameter to pressure derivative of the bulk modulus through following expressions:

$$C_b = \sqrt{\frac{B_0}{\rho_0}} \tag{6}$$

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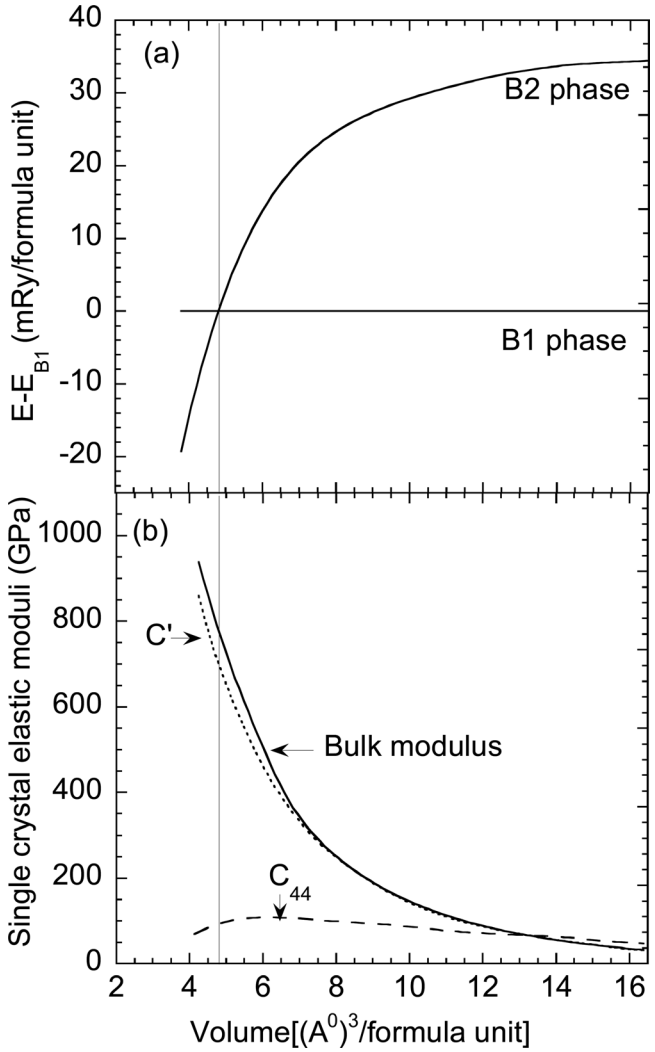


FIG. 1. (a) Total energy of B2 phase relative to that of B1 phase for LiH at various volumes. (b) Elastic moduli of B1 phase as a function of volume.

$$B_0' = 4s - 1. \tag{7}$$

The Greuniesen parameter (γ) is estimated from B_0' using the well-known Dugdale MacDonald relation.[34] These values are also given in Table I. The bulk sound velocity determined theoretically exhibits a good agreement with 6.43 km/s obtained from experimentally measured elastic constants[20] at 140 K. The parameter 's' is within 0.8% of the experimental value of 1.16 from shock data.[35] The value of Gruneisen parameter turns out to be 1.30 as compared to 1.2 determined at 140 K by James and Kheyrandish[20] using experimental value of thermal expansion coefficient, compressibility and heat capacity.

In Table I, we have also compared the theoretically determined elastic constants at ambient pressure with the experimental data measured through ultrasonic technique[20,21] and other theoretical values calculated using a periodic *ab-initio* (HF and DFT) code "CRYSTAL" based on an atom-centered (Gaussian) basis set program[22] and *ab-initio* pseudopotential plane wave method.[18] The presently determined values are closer to the values measured in ultrasonic experiments[20] at the temperature of $\sim$60 K and 140 K.

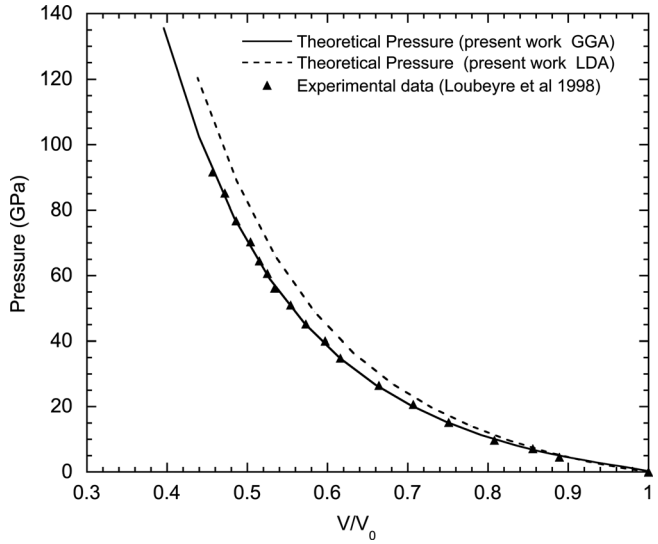


FIG. 2. Comparison of theoretical 300 K isotherm of LiH with experimental data (Ref. 10). The theoretical 300 K isotherm is shown for both GGA and LDA calculations. The experimental data are read from Fig. 3 of Loubeyre *et al.* (Ref. 10). The data contain values for both Li^7H and Li^7D. As pointed out by these authors, the isotherm for the two compounds are almost identical as the correction for zero point motion ranges between 0.3 to 1 GPa only from 0 to 94 GPa.

A further confirmation comes from the calculation of the Debye temperature (θ_0) from these elastic constants. The value of θ_0, turns out to be 828 K, which is near the value 810 K derived through the experimentally measured specific heat.[36]

In Fig. 1(a), we have plotted the total energy of the B2 phase relative to that of the B1 phase for various volumes. As depicted in the figure, our calculations suggest that the B1→B2 phase transition takes place at a volume of 4.8 (A^0)^3/formula unit (V/V$_0$ $\sim$0.29). The corresponding pressure is $\sim$327 GPa. This is in accord with the predictions of Wang *et al.*[2] and Zurek *et al.*[19] and not with Gou *et al.*[17] and Zhang *et al.*[18] The calculated 300 K isotherm is plotted in Fig. 2 along with the experimental data.[10] The agreement between theory and experiment is excellent and is better than with some other theoretical predictions. Further, it is clear from the figure that 300 K isotherm is better described by GGA as compared to LDA.

After finding that our calculated elastic constants at zero pressure and the experimental data are in good agreement, we have performed further calculations for determinations of elastic constants as a function of compression for B1 phase. As shown in Fig. 1(b), the bulk modulus and shear modulus C′ of B1 phase increase monotonically with increasing compression, which is similar to the trend found by Zhang *et al.*[18] However, the shear modulus C$_{44}$ determined from our calculations initially increases monotonically with compression and reaches its maximum value of $\sim$109 GPa at a volume of $\sim$5.97 (A^0)^3/formula unit (corresponding pressure $\sim$196 GPa), thereafter, starts softening upon further compression. However, it does not soften completely at the B1→B2 transition pressure. This softening is unlike that reported by Zhang *et al.*,[18] where C$_{44}$ increases monotonically up to the pressure of 600 GPa. It was predicted by

05 November 2024 18:44:40

TABLE II. Theoretically calculated single crystal elastic constants of B1 phase of LiH at various unit cell volume and corresponding hydrostatic pressures.

Volume (Å^3/formula unit)	Pressure (GPa)	C11 (GPa)	C12(GPa)	C44(GPa)
15.94	0.39	79.9	13.4	49.1
14.87	3.17	107.2	14.2	55.7
14.42	4.67	118.2	16.0	61.4
13.66	7.67	139.3	19.7	65.1
12.90	11.43	166.1	23.4	67.2
12.14	16.12	197.0	28.8	70.4
11.38	22.05	233.9	35.9	73.9
10.62	29.66	281.9	43.7	80.3
9.87	39.64	346.8	50.7	88.5
9.11	52.94	421.8	63.3	93.4
8.35	70.87	528.7	73.7	97.0
7.59	95.14	643.0	98.9	101.4
6.83	127.93	813.7	120.7	107.7
5.97	196.1	1125.8	203.6	109.1
5.16	281.25	1485.1	269.9	104.9
4.85	323.75	1662.4	306.2	99.2

Zhang *et al.*[18] that LiH is different in this respect from NaH, KH and CsH. We do not support this conclusion. The numerical values of elastic constants are shown in Table II. No experimental data are available for comparison at higher pressures. Additionally, Fig. 3 shows the elastic anisotropy $[A = 2C_{44}/(C_{11} - C_{12})]$ parameter as a function of pressure. The parameter "A" at zero pressure is found to be ~ 1.48. As pointed out by James and Kheyrandish,[20] this is the largest among the alkali halides in the NaCl structure. Its value becomes near to that of the heavier alkali halides only at higher pressures.

Finally, the Fig. 4 shows the polycrystalline elastic moduli in the Voiget and Reuss approximation as a function of pressure. As is clear from the figure, G_V and G_R determine the upper and lower bounds of the shear modulus. The Voiget-Reuss- Hill approximated value of shear modulus G which is the average of the G_R and G_V is also shown in the figure. From these, at zero pressure the bulk modulus, shear modulus and the Poisson ratio are 34.2 GPa, 41.7 GPa and 0.14, respectively. G_R follows the trend of C_{44} with pressure.

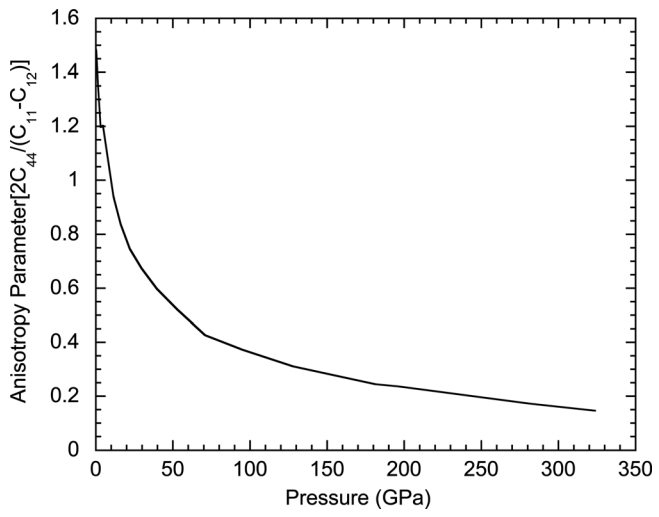


FIG. 3. Theoretically calculated elastic anisotropy of LiH single crystal as a function of pressure.

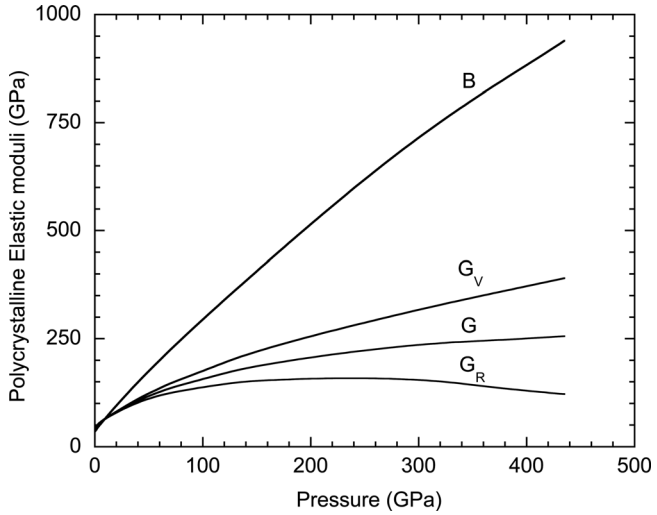


FIG. 4. Isotropic elastic moduli of LiH under Voiget, Reuss and Voiget-Reuss-Hill approximations.

IV. Summary

The structural stability analysis suggests that B1 phase of LiH will transform to B2 phase at a compression of ~ 0.29 with corresponding transition pressure of ~ 327 GPa. The 300 K isotherm determined theoretically agrees closely with experimental P-V data. Various physical quantities derived from theoretical equation of state compare well with the available experimental data. The pressure dependent elastic constants have also been determined. These compare well with the experimental data at ambient pressure.

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